JEE Mathematics: Quick Review with Grading and Practice Map - Trigonometric Equation (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (37 questions; a question that appears in two of these is counted once): High = 3 or more questions, Medium = 1 to 2 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction

Question 01

What do you mean by principal solution of a trigonometric equation?

Model Answer

Principal solution: a solution lying in $[0, 2\pi)$. Particular solution: a solution in a given interval (the principal solution is a special case).

Ex: $\sin\theta = \dfrac{1}{2} \Rightarrow$ principal solutions $\theta = \dfrac{\pi}{6},\ \dfrac{5\pi}{6}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

What do you mean by general solution of a trigonometric equation?

Model Answer

General solution: one formula (with $n \in I$) that gives all values satisfying the equation; it exists because the ratios are periodic.

Ex: $\sin\theta = 0 \Rightarrow \theta = n\pi,\ n \in I$.

Grade - Low

Practice map, main concept: none

Also used in: none

Important Results for standard trig equations

Question 03

What is the formula for general solutions of $\sin x = 0$?

Model Answer

$$\sin x = 0 \Rightarrow x = n\pi,\ n \in I$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 04
What is the formula for general solutions of $\sin x = 1$?

Model Answer

$$\sin x = 1 \Rightarrow x = (4n+1)\frac{\pi}{2} = 2n\pi + \frac{\pi}{2},\ n \in I$$

Grade - **Low**

Practice map, main concept: none
Also used in: Sheet: J-I 16

Question 05
What is the formula for general solutions of $\sin x = -1$?

Model Answer

$$\sin x = -1 \Rightarrow x = (4n-1)\frac{\pi}{2} = 2n\pi - \frac{\pi}{2},\ n \in I$$

Grade - **Low**

Practice map, main concept: none
Also used in: none

Question 06
What is the formula for general solutions of $\cos x = 0$?

Model Answer

$$\cos x = 0 \Rightarrow x = (2n+1)\frac{\pi}{2},\ n \in I$$

Grade - **Low**

Practice map, main concept: none
Also used in: DPP: 7(8)

Question 07
What is the formula for general solutions of $\cos x = 1$?

Model Answer

$$\cos x = 1 \Rightarrow x = 2n\pi,\ n \in I$$

Grade - **Low**

Practice map, main concept: none
Also used in: Sheet: J-I 16

Question 08
What is the formula for general solutions of $\cos x = -1$?

Model Answer

$$\cos x = -1 \Rightarrow x = (2n+1)\pi,\ n \in I$$

Grade - **Low**

Practice map, main concept: none
Also used in: none

General Solutions for trig equations

Question 09

What is the formula for general solutions of $\sin x = \sin\alpha$, $\cos x = \cos\alpha$, $\tan x = \tan\alpha$?

Model Answer

$$\sin x = \sin\alpha \Rightarrow x = n\pi + (-1)^n\alpha; \quad \cos x = \cos\alpha \Rightarrow x = 2n\pi \pm \alpha; \quad \tan x = \tan\alpha \Rightarrow x = n\pi + \alpha$$

Ex: $\sin\theta = \dfrac{\sqrt{3}}{2} \Rightarrow \theta = n\pi + (-1)^n\dfrac{\pi}{3}$; $\sec 2\theta = -\dfrac{2}{\sqrt{3}} \Rightarrow \cos 2\theta = \cos\dfrac{5\pi}{6} \Rightarrow \theta = n\pi \pm \dfrac{5\pi}{12}$.

Grade - High

Practice map, main concept: Sheet: J-M 13, 15; C-I 16 | DPP: 6(1-8, 26); 8(21)

Also used in: Sheet: J-I 1-2, 6-7; J-II 7, 17, 26; J-A 5; C-I 3, 10, 14; C-II 22-23 | DPP: 6(9-11, 18-21, 24, 28-29); 7(6)

Question 10

In which interval should the value of α be taken while applying the formula for general solutions?

Model Answer

Take α as the principal value: sin, tan: $\alpha \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$; cos: $\alpha \in [0, \pi]$.

Ex: $\cos x = -\dfrac{1}{2} \Rightarrow \alpha = \dfrac{2\pi}{3}$ (not $\dfrac{4\pi}{3}$), so $x = 2n\pi \pm \dfrac{2\pi}{3}$.

Grade - Low

Practice map, main concept: Sheet: C-II 20

Also used in: Sheet: C-II 19 | DPP: 6(1-8)

Question 11

What is the formula for general solutions of $\operatorname{cosec} x = \operatorname{cosec}\alpha$, $\sec x = \sec\alpha$, $\cot x = \cot\alpha$?

Model Answer

Same as their reciprocals: cosec like sin $(n\pi + (-1)^n\alpha)$, sec like cos $(2n\pi \pm \alpha)$, cot like tan $(n\pi + \alpha)$.

Ex: $\sec\theta = \sqrt{2} \Rightarrow \theta = 2n\pi \pm \dfrac{\pi}{4}$.

Grade - Low

Practice map, main concept: DPP: 6(9-11)

Also used in: none

Question 12

What is the formula for general solutions of $\sin^2 x = \sin^2\alpha$, $\cos^2 x = \cos^2\alpha$, $\tan^2 x = \tan^2\alpha$?

Model Answer

$$\sin^2 x = \sin^2\alpha, \ \cos^2 x = \cos^2\alpha, \ \tan^2 x = \tan^2\alpha \ \Rightarrow \ x = n\pi \pm \alpha$$

Same for $\cot^2, \sec^2, \operatorname{cosec}^2$. Ex: $4\tan^2\theta = 3\sec^2\theta \Rightarrow \sin^2\theta = \dfrac{3}{4} = \sin^2\dfrac{\pi}{3} \Rightarrow \theta = n\pi \pm \dfrac{\pi}{3}$.

Grade - Low

Practice map, main concept: DPP: 6(12-17, 27)

Also used in: Sheet: J-II 27; C-II 7 | DPP: 7(16-17)

Question 13

How to find solutions in any particular interval with the help of general solutions?

Model Answer

Write the general solution, put $n = 0, \pm1, \pm2, \ldots$ and keep the values inside the interval (or count intersections on the graph). Reject values where the equation is not defined.

Ex: $2\sin^2 x + 5\sin x - 3 = 0 \Rightarrow \sin x = \frac{1}{2}$; in $[0, 3\pi]$: $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$, i.e. 4 solutions.

Grade - High

Practice map, main concept: Sheet: J-II 17; J-A 5; C-I 14; C-II 21 | DPP: 6(20-21)

Also used in: Sheet: J-I 3-4, 8-10, 12, 21; J-II 3-4, 8-9, 16, 20, 23-24; J-M 1-15, 17-19; J-A 2-3; C-I 7-8, 11, 13; C-II 1-2, 9, 11 | DPP: 6(26); 8(39) | Question Bank: 11-14, 16, 19, 22

Question 14

How to check whether two general solutions expressed differently are equal?

Model Answer

Put $n = 0, \pm1, \pm2, \ldots$ in both and compare the values in one period ($[0, 2\pi)$); or split n into even/odd ($n = 2k$, $2k + 1$) and reduce both to the same form.

Ex: $(2n-1)\frac{\pi}{2} + (-1)^n\frac{\pi}{3}$ and $2n\pi \pm \frac{\pi}{6}$ both give $\frac{\pi}{6}, \frac{11\pi}{6}$ in $[0, 2\pi)$: the same angles.

Grade - Low

Practice map, main concept: Sheet: J-I 6; J-II 31; C-II 22 | DPP: 6(22-24)

Also used in: DPP: 6(25)

Types of Trigonometric Equations Type 1

Question 15

What is the basic idea of solving type 1 questions?

Model Answer

Factorisation: bring all terms to one side, write in one ratio (quadratic in $\sin x$, $\cos x$, $\tan x$, ...) or factorise, and solve each factor; reject values outside $[-1, 1]$.

Never cancel a common factor (it may be zero); take it as a separate factor.

Ex: $2\sin^2 x - 3\sin x + 1 = 0 \Rightarrow (2\sin x - 1)(\sin x - 1) = 0 \Rightarrow x = n\pi + (-1)^n\frac{\pi}{6}$ or $x = 2n\pi + \frac{\pi}{2}$.

Grade - High

Practice map, main concept: Sheet: J-I 2-4, 11, 18-19; J-II 1, 23-24, 26-27; J-M 7-10, 12, 19; J-A 2; C-I 2-3, 7-8, 10; C-II 4-5, 7-10, 14 | DPP: 7(1-20); 8(18, 20, 39) | Question Bank: 11-12, 16, 19-20, 22-23

Also used in: Sheet: J-I 20, 22; J-II 6, 10, 19, 25; J-M 17; C-I 16; C-II 3, 12 | DPP: 6(16); 8(38) | Question Bank: 21

Types of Trigonometric Equations Type 2

Question 16

How to solve equation of the type $a\sin x + b\cos x = c$?

Model Answer

Divide by $\sqrt{a^2+b^2}$, put $\dfrac{a}{\sqrt{a^2+b^2}} = \cos\theta$, $\dfrac{b}{\sqrt{a^2+b^2}} = \sin\theta$: $\sin(x+\theta) = \dfrac{c}{\sqrt{a^2+b^2}}$. Solvable only if $|c| \le \sqrt{a^2+b^2}$.

Ex: $\sqrt{3}\sin x - \cos x = \sqrt{2} \Rightarrow \sin\left(x - \dfrac{\pi}{6}\right) = \dfrac{1}{\sqrt{2}} \Rightarrow x = n\pi + (-1)^n\dfrac{\pi}{4} + \dfrac{\pi}{6}$.

Alternative: $t = \tan\dfrac{x}{2}$ gives a quadratic in t (check $x = (2n+1)\pi$ separately); its roots give $\tan\dfrac{\alpha}{2}, \tan\dfrac{\beta}{2}$ for the two roots α, β.

Grade - **High**

Practice map, main concept: Sheet: J-I 5; J-II 9, 11, 15, 25, 28, 32; J-A 4 | DPP: 7(21-26, 28-32) | Question Bank: 15, 17

Also used in: Sheet: J-II 2, 31; J-A 3, 5; C-I 1, 9; C-II 4, 15-16

Question 17

How to assume θ when values other than standard values are obtained?

Model Answer

Let α be the principal value: $\sin x = k \Rightarrow x = n\pi + (-1)^n\sin^{-1}k$; $\cos x = k \Rightarrow 2n\pi \pm \cos^{-1}k$; $\tan x = k \Rightarrow n\pi + \tan^{-1}k$. In $a\sin x + b\cos x = c$ take $\theta = \tan^{-1}\dfrac{b}{a}$ $(a > 0)$.

Ex: $3\sin x + 4\cos x = 5 \Rightarrow \sin(x+\theta) = 1$ with $\tan\theta = \dfrac{4}{3} \Rightarrow x = 2n\pi + \dfrac{\pi}{2} - \tan^{-1}\dfrac{4}{3}$.

Grade - **High**

Practice map, main concept: Sheet: J-M 18 | Question Bank: 14

Also used in: Sheet: J-II 1, 12, 28; C-I 7-8; C-II 8, 17 | DPP: 7(2-3, 10, 15, 20, 25-27); 8(19-20) | Question Bank: 15, 17

Question 18

How to solve equation of the type $a\sin^2 x + b\cos^2 x + c\sin x\cos x = d$?

Model Answer

Write $d = d(\sin^2 x + \cos^2 x)$ and divide by $\cos^2 x$ (check $\cos x = 0$ separately): quadratic in $\tan x$. Or use $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$, $\sin x\cos x = \frac{\sin 2x}{2}$ to get Type 2 in $2x$.

Ex: $1 + \sin^2\theta = 3\sin\theta\cos\theta \Rightarrow 2\tan^2\theta - 3\tan\theta + 1 = 0 \Rightarrow \tan\theta = 1,\ \dfrac{1}{2}$.

Grade - **Low**

Practice map, main concept: DPP: 7(27)

Also used in: Sheet: J-II 18

Types of Trigonometric Equations Type 3

Question 19

How to solve equations of the type $\sin x = \sin\alpha$, $\cos x = \cos\alpha$, $\tan x = \tan\alpha$ when α is also variable?

Model Answer

Use the same general formula with α a function of x, then solve for x; for $(-1)^n$ split $n = 2k$ and $n = 2k+1$. Change cos to sin (or cot to tan) by complementary angles first.

Ex: $\sin 3x = \sin x \Rightarrow 3x = n\pi + (-1)^n x$: $n = 2k \Rightarrow x = k\pi$; $n = 2k+1 \Rightarrow x = (2k+1)\dfrac{\pi}{4}$.

Grade - **Medium**

Practice map, main concept: Sheet: J-II 2, 5; J-M 3; J-A 3; C-I 1, 9; C-II 1, 16, 19 | DPP: 6(25); 8(14-15, 17, 24-38)
Also used in: Sheet: J-II 13, 22, 28; C-II 20-21

Question 20

What is the basic idea of solving type 3 questions?

Model Answer

Convert sums to products ($C \pm D$) or products to sums ($2\sin A\cos B$, ...) so that the equation factorises; pair terms whose angles add or differ equally.

Ex: $\sin\theta + \sin 5\theta = \sin 3\theta \Rightarrow \sin 3\theta(2\cos 2\theta - 1) = 0 \Rightarrow \theta = \dfrac{n\pi}{3}$ or $n\pi \pm \dfrac{\pi}{6}$.

Grade - **High**

Practice map, main concept: Sheet: J-I 7-10; J-II 3-4, 6, 16; J-M 2, 5; C-II 11, 23 | DPP: 8(1-13, 16, 19, 22) | Question Bank: 13
Also used in: Sheet: J-II 5; C-I 19; C-II 5 | DPP: 8(21)

Types of Trigonometric Equations Type 4

Question 21

How to use substitution in trigonometric equations?

Model Answer

Spot one expression that the whole equation can be written in ($\sin 2x$, $\sin^2 x$, $\tan\frac{x}{2}$, $a^{\sin^2 x}$, $\tan x + \cot x$, ...), put it $= t$, solve the algebraic equation and keep only t in its range.

Ex: $\sin^4 x + \cos^4 x = \sin x\cos x$, $t = \sin 2x$: $1 - \dfrac{t^2}{2} = \dfrac{t}{2} \Rightarrow t = 1 \Rightarrow x = n\pi + \dfrac{\pi}{4}$.

Grade - **High**

Practice map, main concept: Sheet: J-II 7, 10; J-M 4, 6, 14; C-II 2, 12-13 | Question Bank: 21
Also used in: Sheet: J-II 21, 32; J-A 4

Question 22

Which expression to put t when both $(\sin x + \cos x)$ and $\sin 2x$, or $(\sin x - \cos x)$ and $\sin 2x$, are present in the equation?

Model Answer

$$t = \sin x + \cos x \Rightarrow \sin 2x = t^2 - 1; \qquad t = \sin x - \cos x \Rightarrow \sin 2x = 1 - t^2; \qquad |t| \le \sqrt{2}$$

Ex: $\sin x + \cos x = 1 - \sin 2x \Rightarrow t^2 + t - 2 = 0 \Rightarrow t = 1$ ($t = -2$ rejected) $\Rightarrow x = 2n\pi$ or $2n\pi + \dfrac{\pi}{2}$.

Grade - **Low**

Practice map, main concept: none
Also used in: none

Types of Trigonometric Equations Type 5

Question 23

How would you know whether an equation is of type 5?

Model Answer

Boundedness: the equation cannot be reduced by the usual methods, but one side has a maximum equal to the other side's minimum, e.g. sum of sines $= 2$, product $= \pm 1$, $\sin^m x + \cos^n x = 1$, trig $=$ algebraic expression ($x^2 - 2x + 3$, $2^x + 2^{-x}$), two variables in one equation.

Ex: $\sin^{10} x + \cos^{11} x = \dfrac{5}{4}$ has no solution, since $\sin^m x + \cos^n x \leq 1$ for $m, n \geq 2$.

Grade - High

Practice map, main concept: Sheet: J-I 17

Also used in: Sheet: J-I 13-15; J-II 8; J-M 1, 16; J-A 1; C-I 4-5; C-II 6 | Question Bank: 18

Question 24

If a question is of type 5 then how to solve that equation?

Model Answer

Find the range of each side; equality forces every term to take its extreme value at the same x. Solve each condition and take the common values (match n and m by equating the two general solutions).

Ex: $\sqrt{3}\sin x + \cos x = y^2 + 2$: LHS $\leq 2 \leq$ RHS $\Rightarrow y = 0$, $\cos\left(x - \dfrac{\pi}{3}\right) = 1 \Rightarrow x = 2n\pi + \dfrac{\pi}{3}$.

Grade - High

Practice map, main concept: Sheet: J-I 12-16; J-II 8, 20; J-M 1, 11, 16; J-A 1; C-I 4-5, 11, 15, 17-18; C-II 6, 18 | Question Bank: 18

Also used in: Sheet: J-I 17-19; J-II 30; C-I 2 | Question Bank: 20, 23

Types of Trigonometric Equations Type 6

Question 25

What to keep in mind if we do squaring in trigonometric equations?

Model Answer

Squaring adds extraneous roots (it also solves the equation with the opposite sign). Avoid it if possible; otherwise substitute every root in the original equation. Also reject roots where the equation is not defined ($\tan, \sec$ with $\cos x = 0$, a log base, a square root).

Ex: $\sin\theta + \cos\theta = 1 \Rightarrow \sin 2\theta = 0 \Rightarrow \theta = \dfrac{n\pi}{2}$; but $\pi, \dfrac{3\pi}{2}$ give -1, so only $\theta = 2n\pi,\ 2n\pi + \dfrac{\pi}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 12-14; C-I 13

Also used in: Sheet: J-I 2-3, 5; J-II 16, 31; J-M 7; C-II 14 | DPP: 7(28-29, 31-32)

Question 26

How to write general solution if we have done squaring in a trigonometric equation?

Model Answer

Find the solutions in one period $[0, 2\pi)$, keep those that satisfy the original equation (correct signs of $\sin x, \cos x$), then add $2n\pi$ to each.

Ex: $2\sin\left(3x + \dfrac{\pi}{4}\right) = \sqrt{1 + 8\sin 2x \cos^2 2x} \Rightarrow x = 2n\pi + \dfrac{\pi}{12}$ or $2n\pi + \dfrac{17\pi}{12}$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 11-15

Trigonometric Inequalities

Question 27

How to find general solutions of basic trigonometric inequalities?

Model Answer

Solve in one period (for $\sin, \cos$ a 2π interval, for $\tan, \cot$ a π interval) using the boundary equation, then add $2n\pi$ (or $n\pi$).

Ex: $\sin x > \dfrac{1}{2} \Rightarrow x \in \left(2n\pi + \dfrac{\pi}{6},\ 2n\pi + \dfrac{5\pi}{6}\right)$.

Grade - Low

Practice map, main concept: Sheet: J-I 20; J-II 18, 30; C-I 12; C-II 15

Also used in: Sheet: C-I 20

Question 28

How to solve trigonometric inequalities with the help of graph?

Model Answer

Reduce to one ratio of one angle ($\sin y < k$, ...), draw $y = \sin$ and the line $y = k$ over one period, read where the curve is above/below, then substitute back and add the period.

Ex: $\sqrt{3}\sin x + \cos x < 1 \Rightarrow \sin\left(x + \dfrac{\pi}{6}\right) < \dfrac{1}{2} \Rightarrow x \in \left(2n\pi - \dfrac{4\pi}{3},\ 2n\pi\right)$.

Grade - Low

Practice map, main concept: Sheet: C-I 20

Also used in: Sheet: C-I 12

System of trigonometric equations

Question 29

How to solve system of 2 or more trigonometric equations?

Model Answer

One variable: solve one equation in $[0, 2\pi)$, keep the values that satisfy the other, add $2n\pi$. Two variables: add/subtract or divide to get equations in $x + y$ and $x - y$, solve each with different integers n, m, then solve for x, y.

Ex: $\cos x \cos y = \dfrac{3}{4},\ \sin x \sin y = \dfrac{1}{4} \Rightarrow \cos(x - y) = 1,\ \cos(x + y) = \dfrac{1}{2} \Rightarrow x = (n + m)\pi \pm \dfrac{\pi}{6},\ y = (m - n)\pi \pm \dfrac{\pi}{6}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 1, 21-22; J-II 19, 21-22; J-M 17; C-I 19; C-II 3, 17 | DPP: 6(18-19, 28-29)

Also used in: Sheet: J-II 20; C-I 15, 17-18; C-II 18